

Building Sampling Distributions with StatKey

StatKey · Lock5 / Statkey

Year 10

~30 minutes

Mathematics · Statistics

NOTICE — AI-generated worksheet

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Simulation: <https://www.lock5stat.com/StatKey/>

Curriculum links: AC9M10P02 · AC9M10ST02 · AC9M10ST01

Learning intentions

- I can use StatKey to generate a bootstrap distribution from sample data.
- I can describe the shape, centre and spread of a simulated sampling distribution.
- I can use simulation results to comment on the reliability of a statistical claim.

Getting started

Open StatKey and choose one of the built-in datasets under 'Bootstrap Confidence Intervals' (e.g. a dataset for a single mean). Look carefully at the original sample data displayed before you generate any resamples.

Tasks

1. Describe the shape of the original sample's distribution shown before you generate any resamples.

2. Click 'Generate 1 Sample' several times. What changes each time, and what stays roughly the same?

3. Generate resamples in increasing batches (e.g. 1, then 100, then 1000) and describe the shape of the bootstrap distribution at each stage.

Number of resamples generated	Shape of bootstrap distribution

4. Based on your table, explain what happens to the bootstrap distribution as you increase the number of resamples, and why this matters for estimating a confidence interval.

5. Use the two-tail option to generate a 95% confidence interval. Record the interval StatKey gives you.

6. Explain, in your own words, what this 95% confidence interval tells you about the population — and what it does NOT tell you.

7. Switch to a randomization test using one of StatKey's built-in comparison datasets. State the claim being tested in your own words.

8. A newspaper article claims a difference between two groups is 'significant'. Using what StatKey has shown you about randomization distributions, explain what evidence you would want to see before accepting that claim.

Extension (optional)

Choose a different built-in dataset and repeat the bootstrap confidence interval process. Compare the width of this new interval to your first one, and suggest one reason the two intervals might differ.
